



ERGONOMICS AND YOUR PC

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Personal computers have been described as mind tools. Using one should thus be a comfortable experience with the minimum of distractions to interfere with the flow of creativity. Even the most mind-numbing of data-entry jobs can be made easier through using the right input devices in a comfortably set up working space.

Ergonomics can be defined as 'design for ease of human use'. Of course other design factors tend to take priority. For example there's engineering practicality — "let's just get the thing to work first, figure out how to produce it simply and we'll worry about a designer case later". Or ergonomics — "moulding a contoured handle would cost how much; no, let's just stick a round handle on it". Apple is a good example of a computer company that has made a virtue out of design ergonomics.

Why are ergonomics often ignored in the workplace?

Applying good ergonomic principles takes care and effort and may even cost money. Employers perhaps have trouble recognising the benefits from having a more productive and healthier workforce. Supplying workers with a desk, a chair and a computer takes little thought. Ensuring they have a comfortable and productive work environment takes effort; indeed employers may think "why bother". In many countries legislation attempts to protect workers by dictating, or at least recommending, good working practice.

Several of the major IT equipment manufacturers publish safety and comfort guides for computer users. Hewlett Packard for example makes its [ref.1] Safety & Comfort Guide available as a PDF to download and IBM and Lenovo publish [ref.2] a guide to Workstation ergonomics on their web site.

A search of the official Indian government web site — india.gov.in — for computer ergonomics reveals little, other than some notes on the comfort of tractor seats. However, India does have one professional body for ergonomists, the [ref.3] Indian Society of Ergonomics, established in 1983. According to its web site the ISE is the only professional body representing ergonomics / human factors professionals in India. It publishes a quarterly newsletter.

The display

It is not uncommon to see users elevating their display height by placing it on stacks of books or other makeshift stands. In an office setting this can give an unprofessional impression and may be unstable and hazardous. This behaviour is hard to explain and is, ergonomically speaking, incorrect. In any case most recent LCD monitors have built-in height adjustment. The correct height for a computer display is for the top edge of the screen to be at, or just a little below, the users seated eye line.

Another bad habit is festooning the edges and top of a display with trinkets and

post-it notes. Again this looks unprofessional and can cause stress. Odd brightly coloured objects around the edge of a screen attract the eye and trying to sub-consciously ignore them, while you concentrate on what's on the screen, can cause physical strain and mental tension. If a workstation is used for any aspect of design or image manipulation, then the surrounding colours of walls, ceilings and nearby objects and the type of lighting are even more critical. In this case anything other than neutral colours and 'daylight' lighting will distort the perception of on-screen colours.

Viewing distance does depend somewhat on screen size, but 50 to 60 cm is a normal viewing distance.

Lighting

All too often computers are set up in environments where there is no control over, or thought given to, either artificial or natural lighting. A modern office building should be designed with computer use in mind, but usually they aren't. Ideally, computer screens should be arranged so that no strong artificial or natural light source directly reflects from their surface. Sitting either facing or with your back to a window means that either the natural light behind the screen is too bright, or the light falling on the screen overwhelms the screen image. If your work position is near a window it's best to sit with the window to one side and to place the back of the screen towards and at an angle to the window.

If nearby artificial lighting causes a strong reflection on your screen that cannot be solved by changing the displays position, then a shade over the light source or even removal of the lamp should be considered.

For design environments 'Daylight' classified light sources should be used, usually either D50 or D65. Where workplaces are illuminated with fluorescent tubes this can just be a case of replacing the tubes with D65 tubes.

External light filters that can be placed over the screen are available, but these are often quite ineffective, it's better to deal with the sources of glare and reflection than to try and use a filter to reduce them. Current LCD monitors with a flat, matte surface screen, are less prone to reflections than the older CRT displays and often easier to manoeuvre into a suitable position.

The keyboard

The keyboard has been and still is the main human input device. Early computers used the electro-mechanical teletype. The teletype design was in turn based on the typewriter, a device that is now 124 years old. The odd QWERTY key arrangement is often commented on. The common myth is that the designers of the typewriter, originally a slow mechanical device, were forced to jumble up the order of the letters, in an effort to slow down the typist to the point where the letters no longer jammed as several tried to strike the paper at almost the same time. The truth is apparently the other way about, the original designer realised that key jams where more likely to occur between common letter pairs, so he arranged these letters so their mechanical linkages were far apart, reducing jams and therefore increasing the maximum possible typing speed.

In 1932 Prof. August Dvorak, of Washington State University set out to design a better keyboard layout, with more of the most commonly used letters (in English) in the 'home row'. Dvorak keyboards turned out not to be as obviously efficient, or as popular, as Prof. Dvorak hoped and today the venerable QWERTY layout keyboard is still with us. The [ref.4] argument over which keyboard layout is better continues to this day...



A teletype, forerunner of today's computer keyboards. The round key design echoes that of the older typewriters